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JEE Main Physics Practice Sheet — DPP-049

Topic: Motion of Centre of Mass; Problems & Dynamics

Time Allowed: 60 Minutes — Max Marks: 80 — Marking Scheme: +4, -1

- Q1.** A projectile of mass M is fired from the ground with speed u at an angle θ to the horizontal. At the highest point of its trajectory, it explodes into two fragments of masses $M/3$ and $2M/3$. If the lighter fragment falls vertically downward immediately after the explosion, the trajectory of the centre of mass of the system until the first fragment hits the ground is:
- (A) A vertical straight line
(B) Exactly the original parabolic path
(C) Shifted towards the heavier fragment
(D) A horizontal straight line
- Q2.** Two blocks of masses $m_1 = 2$ kg and $m_2 = 4$ kg connected by a light spring are placed on a smooth horizontal floor. A constant horizontal force $F = 18$ N acts on the 4 kg block. The acceleration of the centre of mass of the system is:
- (A) 4.5 m/s^2
(B) 2.0 m/s^2
(C) 3.0 m/s^2
(D) 1.5 m/s^2
- Q3.** A man of mass $M = 60$ kg stands at one end of a boat of mass $m = 120$ kg and length $L = 6$ m floating on calm water. The man walks from one end to the other and stops. Neglecting water resistance, the displacement of the boat relative to the water is:
- (A) 3.0 m
(B) 1.5 m
(C) 4.0 m
(D) 2.0 m
- Q4.** Two particles A and B initially at rest move towards each other under a mutual force of attraction. At an instant when the speed of A is v and the speed of B is $2v$, the speed of the centre of mass of the system is:
- (A) 0
(B) v
(C) $1.5v$
(D) $3v$
- Q5.** A body of mass 10 kg is projected with an initial velocity $\vec{v}_0 = (30\hat{i} + 40\hat{j}) \text{ m/s}$. At $t = 2$ s, the body explodes into two pieces of masses 4 kg and 6 kg. The position vector of the centre of mass of the system at $t = 3$ s is ($g = 10 \text{ m/s}^2$):
- (A) $(60\hat{i} + 75\hat{j}) \text{ m}$
(B) $(90\hat{i} + 75\hat{j}) \text{ m}$
(C) $(90\hat{i} + 120\hat{j}) \text{ m}$
(D) $(60\hat{i} + 90\hat{j}) \text{ m}$
- Q6.** A block of mass m is placed on the top of a smooth triangular wedge of mass M , height h , and base length b , which rests on a smooth horizontal floor. When the block slides down from the top to the base of the wedge, the horizontal displacement of the wedge is:
- (A) $\frac{mb}{M+m}$
(B) $\frac{Mb}{M+m}$
(C) $\frac{mh}{M+m}$
(D) $\frac{(M-m)b}{M+m}$
- Q7.** Two particles of masses 1 kg and 3 kg are projected vertically upwards from the ground with speeds 40 m/s and 20 m/s respectively. The acceleration of their centre of mass while both are in flight is ($g = 10 \text{ m/s}^2$):
- (A) 5 m/s^2 downwards
(B) 10 m/s^2 downwards
(C) 0
(D) 10 m/s^2 upwards
- Q8.** A circular disc of mass M and radius R is resting on a smooth horizontal surface. A person of mass m standing at the edge of the disc walks along its circumference through an angle of 2π radians. The angle through which the disc rotates during this process is:
- (A) $\frac{2\pi m}{M+2m}$
(B) $\frac{2\pi m}{M+m}$
(C) $\frac{4\pi m}{M+2m}$
(D) $\frac{2\pi M}{M+2m}$
- Q9.** Two blocks of masses m and $2m$ are placed on a smooth horizontal surface and connected by an ideal spring. The system is given an initial compression x_0 and released from rest. At the instant the spring returns to its natural length, the ratio of the kinetic energy of mass m to that of mass $2m$ is:
- (A) 1 : 2
(B) 1 : 4
(C) 4 : 1
(D) 2 : 1

Q10. A uniform chain of length L and mass M is held vertically with its lower end just touching a table and released from rest. At an instant when length y of the chain has fallen on the table, the acceleration of the centre of mass of the chain is:

- (A) g
- (B) $g\left(1 - \frac{y}{L}\right)$
- (C) $g\left(1 - \frac{y^2}{L^2}\right)$
- (D) $g\left(\frac{y}{L}\right)$

Q11. A dumbbell made of two equal masses m connected by a light rigid rod of length l rests on a frictionless horizontal plane. A sharp horizontal impulse J is applied perpendicular to the rod at one of the masses. The speed of the centre of mass of the dumbbell immediately after the impulse is:

- (A) $\frac{J}{m}$
- (B) $\frac{2J}{m}$
- (C) $\frac{J}{2m}$
- (D) $\frac{J}{4m}$

Q12. A shell fired from a cannon explodes mid-air into several fragments of various masses. If the fragments fly off in different directions, the center of mass of all the fragments continues along:

- (A) A straight line in the direction of the heaviest fragment
- (B) The original parabolic path under gravity until external forces change
- (C) A vertical downward path
- (D) A path determined by the kinetic energy released

Q13. Two identical carts A and B , each of mass M , are at rest on smooth horizontal rails. A person of mass m stands on cart A . The person jumps from cart A to cart B with horizontal speed u relative to cart A . The final velocity of cart A is:

- (A) $\frac{mu}{M+m}$ in the opposite direction
- (B) $\frac{mu}{M}$ in the opposite direction
- (C) $\frac{Mu}{M+m}$ in the opposite direction
- (D) $\frac{mu}{2M+m}$ in the opposite direction

Q14. A system consists of three particles of masses 1 kg, 2 kg, and 3 kg. Their position vectors at time t are $\vec{r}_1 = t^2\hat{i}$, $\vec{r}_2 = 2t\hat{j}$, and $\vec{r}_3 = (t^2\hat{i} - t\hat{j})$ respectively. The velocity of the centre of mass of the system at $t = 1$ s is:

- (A) $\frac{4}{3}\hat{i} + \frac{1}{6}\hat{j}$

- (B) $\frac{4}{3}\hat{i} + \frac{1}{3}\hat{j}$
- (C) $\frac{2}{3}\hat{i} + \frac{1}{6}\hat{j}$
- (D) $\frac{8}{3}\hat{i} + \frac{1}{6}\hat{j}$

Q15. Two bodies of masses m_1 and m_2 are connected by a light string passing over a frictionless light pulley fixed to a ceiling (Atwood machine). If $m_1 > m_2$, the acceleration of the centre of mass of the two-body system is:

- (A) $\left(\frac{m_1 - m_2}{m_1 + m_2}\right)g$
- (B) $\left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2 g$
- (C) $\left(\frac{m_1 + m_2}{m_1 - m_2}\right)g$
- (D) 0

Q16. A person of mass m is climbing up a light vertical rope hanging from a stationary hot-air balloon of mass M in calm air. If the person climbs up with speed v relative to the rope, the absolute upward speed of the balloon relative to the ground is:

- (A) $\frac{mv}{M+m}$ downwards
- (B) $\frac{mv}{M+m}$ upwards
- (C) $\frac{Mv}{M+m}$ downwards
- (D) $\frac{mv}{M}$ downwards

Q17. Two identical masses m are connected by a spring of stiffness k . Mass 1 is pushed against a rigid vertical wall while mass 2 is pulled to compress the spring by x_0 . When mass 2 is released, the velocity of the centre of mass of the system when mass 1 leaves the wall is:

- (A) $\frac{x_0}{2}\sqrt{\frac{k}{m}}$
- (B) $x_0\sqrt{\frac{k}{2m}}$
- (C) $x_0\sqrt{\frac{k}{m}}$
- (D) $\frac{x_0}{4}\sqrt{\frac{k}{m}}$

Q18. A wooden block of mass M resting on a smooth horizontal table is struck by a bullet of mass m moving with speed v . The bullet emerges from the block with speed $v/3$. The velocity of the centre of mass of the (block + bullet) system after the collision is:

- (A) $\frac{2mv}{3(M+m)}$
- (B) $\frac{mv}{M+m}$
- (C) $\frac{mv}{3(M+m)}$
- (D) $\frac{Mv}{M+m}$

Q19. A uniform rod of mass M and length L lies on a smooth horizontal table. A particle of mass m moving with velocity v_0 perpendicular to the rod hits one end of the rod and sticks to it. The velocity of the centre of mass of the combined system after the collision is:

- (A) $\frac{mv_0}{M+m}$
- (B) $\frac{Mv_0}{M+m}$
- (C) $\frac{2mv_0}{M+2m}$

(D) $\frac{mv_0}{M+2m}$

Q20. Two particles of masses m_1 and m_2 are moving with velocities $\vec{v}_1$ and $\vec{v}_2$. If an external force $\vec{F}_{\text{ext}}$ acts on the system, the rate of change of total linear momentum of the system is equal to:

- (A) $(m_1 + m_2)\vec{a}_{\text{cm}}$
- (B) $\vec{F}_{\text{ext}}$
- (C) Both (A) and (B)
- (D) Neither (A) nor (B)

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-049)

Motion of Centre of Mass; Problems & Dynamics

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	B	6	A	11	C	16	A
2	C	7	B	12	B	17	A
3	D	8	C	13	A	18	B
4	A	9	D	14	A	19	A
5	B	10	B	15	B	20	C

Detailed Step-by-Step Solutions

Sol 1. Option (B)

The explosion forces are internal forces between the fragments. The only external force acting on the system is uniform gravity $\vec{F}_{\text{ext}} = M\vec{g}$.

Since $\vec{a}_{\text{cm}} = \vec{g}$, the acceleration of the centre of mass remains unchanged, and its trajectory continues to be exactly the original parabolic path until external influences (such as landing) intervene.

Sol 2. Option (C)

The spring force between the two blocks is an internal action-reaction force. The net external force acting on the two-block system is $F = 18 \text{ N}$.

Acceleration of the centre of mass:

$$a_{\text{cm}} = \frac{F_{\text{ext}}}{m_1 + m_2} = \frac{18}{2 + 4} = \frac{18}{6} = 3.0 \text{ m/s}^2$$

Sol 3. Option (D)

Since no external horizontal force acts on the (man + boat) system, the centre of mass remains horizontally stationary ($\Delta x_{\text{cm}} = 0$).

Let x be the backward displacement of the boat relative to the water:

$$\begin{aligned} M(L - x) - mx &= 0 \\ (M + m)x &= ML \\ x &= \frac{ML}{M + m} = \frac{60 \times 6}{60 + 120} = 2.0 \text{ m} \end{aligned}$$

Sol 4. Option (A)

The mutual attraction force is strictly internal ($\vec{F}_{\text{int}}$) and the initial state of the system is rest ($\vec{v}_{\text{cm}}(0) = \vec{0}$).

Since $\sum \vec{F}_{\text{ext}} = \vec{0}$, we have $\vec{a}_{\text{cm}} = \vec{0}$, meaning the velocity of the centre of mass remains zero throughout the motion:

$$v_{\text{cm}} = 0$$

Sol 5. Option (B)

The explosion occurs due to internal forces, so the motion of the centre of mass is governed solely by gravity with initial velocity $\vec{v}_0 = (30\hat{i} + 40\hat{j}) \text{ m/s}$ and acceleration $\vec{a} = -10\hat{j} \text{ m/s}^2$.

At $t = 3 \text{ s}$:

$$\begin{aligned} x_{\text{cm}} &= v_{0x}t = 30 \times 3 = 90 \text{ m} \\ y_{\text{cm}} &= v_{0y}t - \frac{1}{2}gt^2 = 40(3) - \frac{1}{2}(10)(3)^2 \\ &= 120 - 45 = 75 \text{ m} \end{aligned}$$

Thus, $\vec{r}_{\text{cm}} = (90\hat{i} + 75\hat{j}) \text{ m}$.

Sol 6. Option (A)

No external horizontal force acts on the (wedge + block) system.

Let x be the displacement of the wedge to the left as the block moves a horizontal distance $(b - x)$ to the right relative to ground:

$$m(b - x) - Mx = 0 \implies x = \frac{mb}{M + m}$$

Sol 7. Option (B)

While both particles are in the air, the only external force acting on each particle is gravity (m_1g and m_2g downward). Acceleration of COM:

$$a_{\text{cm}} = \frac{m_1g + m_2g}{m_1 + m_2} = g = 10 \text{ m/s}^2 \text{ downwards}$$

Sol 8. Option (C)

Let θ be the angle turned by the disc. No external torque acts about the central vertical axis of the disc.

Conserving angular momentum about the centre of the disc:

$$\begin{aligned} I_{\text{disc}}\omega_{\text{disc}} &= I_{\text{person}}\omega_{\text{person}} \\ \left(\frac{1}{2}MR^2\right)\theta &= (mR^2)(2\pi - \theta) \\ \frac{1}{2}M\theta + m\theta &= 2\pi m \\ \theta &= \frac{2\pi m}{\frac{1}{2}M + m} = \frac{4\pi m}{M + 2m} \end{aligned}$$

Sol 9. Option (D)

Since the system is released from rest on a smooth surface, total linear momentum is conserved:

$$p_1 = p_2 = p$$

The kinetic energy is $K = \frac{p^2}{2m}$.

Ratio of kinetic energies:

$$\frac{K_1}{K_2} = \frac{p^2/(2m)}{p^2/(2(2m))} = \frac{2m}{m} = \frac{2}{1}$$

Sol 10. Option (B)

Length of chain on table is y (at rest); length remaining in air is $(L - y)$, falling with acceleration g .

Total downward external force acting on the chain equals the weight of the suspended portion:

$$F_{\text{ext}} = \left(\frac{M}{L}\right)(L - y)g$$

Acceleration of the centre of mass:

$$a_{\text{cm}} = \frac{F_{\text{ext}}}{M} = g \left(1 - \frac{y}{L}\right)$$

Sol 11. Option (C)

By impulse-momentum theorem for the whole system:

$$J = P_{\text{total}} = (2m)v_{\text{cm}} \implies v_{\text{cm}} = \frac{J}{2m}$$

Sol 12. Option (B)

Internal forces during explosion cancel out in pairs. As long as external force is solely gravity, $\vec{F}_{\text{ext}} = M_{\text{total}}\vec{g}$, so the centre of mass traces the undisturbed parabolic trajectory.

Sol 13. Option (A)

Let cart A acquire backward speed V_A relative to the ground. Velocity of the person relative to the ground is $(u - V_A)$. Horizontal momentum conservation for cart A and person:

$$m(u - V_A) - MV_A = 0 \\ (M + m)V_A = mu \implies V_A = \frac{mu}{M + m}$$

Sol 14. Option (A)

Individual particle velocities are:

$$\vec{v}_1 = \frac{d\vec{r}_1}{dt} = 2t\hat{i} = 2\hat{i} \text{ m/s} \\ \vec{v}_2 = \frac{d\vec{r}_2}{dt} = 2\hat{j} \text{ m/s} \\ \vec{v}_3 = \frac{d\vec{r}_3}{dt} = (2t\hat{i} - \hat{j}) = (2\hat{i} - \hat{j}) \text{ m/s}$$

Velocity of centre of mass at $t = 1$ s:

$$\vec{v}_{\text{cm}} = \frac{1(2\hat{i}) + 2(2\hat{j}) + 3(2\hat{i} - \hat{j})}{1 + 2 + 3} \\ = \frac{8\hat{i} + \hat{j}}{6} = \frac{4}{3}\hat{i} + \frac{1}{6}\hat{j}$$

Sol 15. Option (B)

Magnitude of acceleration of each block:

$$a = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)g$$

Mass m_1 moves downwards ($\vec{a}_1 = -a\hat{j}$) and m_2 moves upwards ($\vec{a}_2 = +a\hat{j}$).

Acceleration of centre of mass:

$$a_{\text{cm}} = \frac{m_1(-a) + m_2(a)}{m_1 + m_2} = -\left(\frac{m_1 - m_2}{m_1 + m_2}\right)a \\ a_{\text{cm}} = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2 g \quad (\text{downwards})$$

Sol 16. Option (A)

Initially, the system is stationary. No external vertical force acts on the (balloon + person) system ($\sum F_{\text{ext}} = 0$). Let the balloon move downward with speed V relative to ground.

Upward speed of the person relative to ground is $(v - V)$.

By vertical momentum conservation:

$$m(v - V) - MV = 0 \implies V = \frac{mv}{M + m} \text{ downwards}$$

Sol 17. Option (A)

When mass 1 leaves the wall, the spring is at its natural length (zero elastic potential energy).

By conservation of energy for mass 2:

$$\frac{1}{2}kx_0^2 = \frac{1}{2}mv_2^2 \implies v_2 = x_0\sqrt{\frac{k}{m}}$$

At this instant, mass 1 just begins to move ($v_1 = 0$).

Velocity of the centre of mass:

$$v_{\text{cm}} = \frac{m(0) + mv_2}{2m} = \frac{v_2}{2} = \frac{x_0}{2}\sqrt{\frac{k}{m}}$$

Sol 18. Option (B)

Since no external horizontal force acts on the table, the total horizontal momentum of the (block + bullet) system is conserved throughout the collision:

$$P_{\text{total}} = mv$$

Velocity of the centre of mass:

$$v_{\text{cm}} = \frac{P_{\text{total}}}{M + m} = \frac{mv}{M + m}$$

Sol 19. Option (A)

Total initial linear momentum along the direction of motion is $P_i = mv_0$.

Since the collision takes place on a smooth table, no external horizontal force acts.

Velocity of the combined centre of mass:

$$v_{\text{cm}} = \frac{P_i}{M + m} = \frac{mv_0}{M + m}$$

Sol 20. Option (C)

Newton's second law for a system of particles states that $\vec{F}_{\text{ext}} = \frac{d\vec{P}}{dt} = M\vec{a}_{\text{cm}}$, where $M = m_1 + m_2$.

Therefore, both expressions $(m_1 + m_2)\vec{a}_{\text{cm}}$ and $\vec{F}_{\text{ext}}$ represent the rate of change of total linear momentum.